UNIT 5

Toughness Testing, Fatigue Crack Growth, J-Integral, FAD, Mixed Mode, Arrest & Repair

K_{Ic} test | Paris & sigmoidal curve | closure | models | J | FAD | mixed mode | repair

1. Fracture Toughness Testing

Standard	Measures	Specimen
ASTM E399	Plane-strain K_{Ic} (LEFM)	CT, SENB (also arc, disc)
ASTM E561	K–R curve (thin sheet)	M(T), CT
ASTM E1820	J_{Ic}, J–R, CTOD δ (EPFM)	CT, SENB
BS 7448 / ISO 12135	CTOD δ_c, J, K_{Ic}	SENB, CT
ASTM E23	Charpy impact energy, DBTT	10×10 mm V-notch bar
ASTM E647	Fatigue crack growth da/dN–ΔK	CT, M(T)

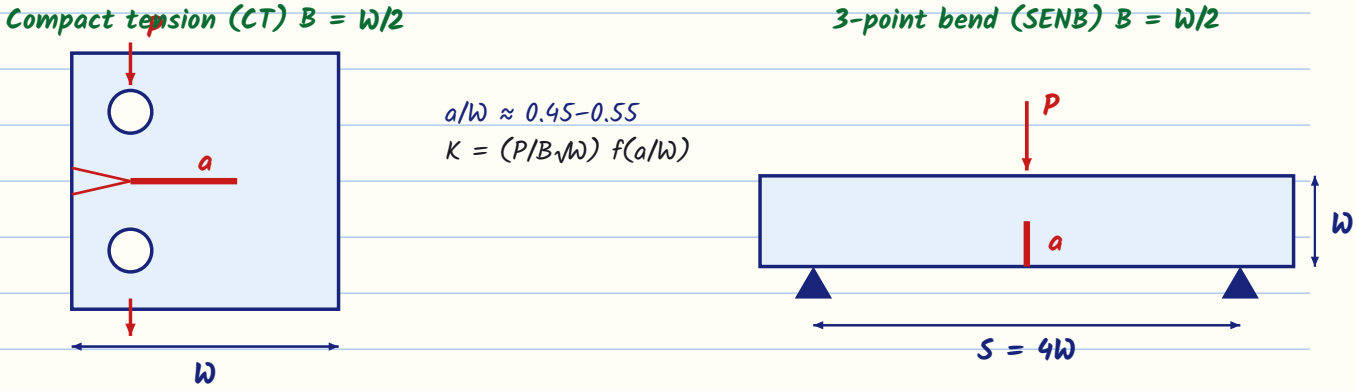
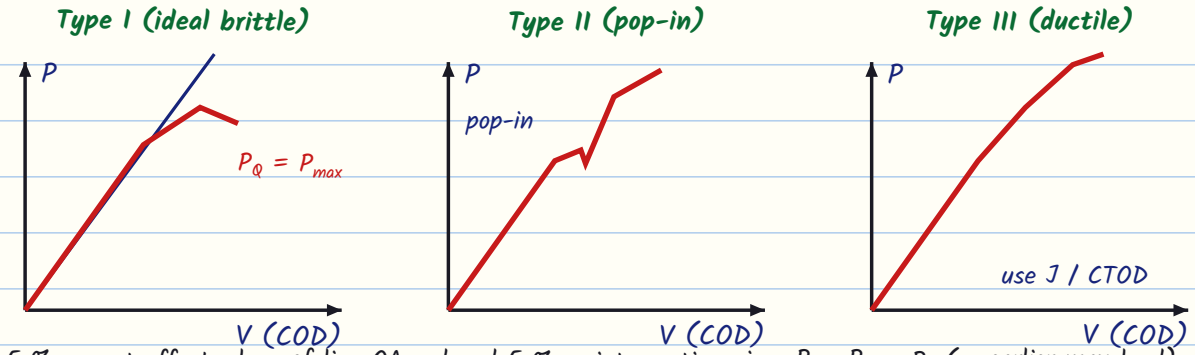


Fig 5.1 Standard K_{Ic} specimens

1.1 E399 procedure

1. Machine specimen: W = 2B, a/W ≈ 0.5 (0.45–0.55), notch with chevron / slot, orientation noted (L-T, T-L).
2. **Fatigue pre-crack** to sharp crack: final $K_{max} \le 0.6\ K_Q$ (to avoid plasticity), ΔK small, length ≥ 1.3 mm.
3. Load at constant displacement rate (K rate 0.55–2.75 MPa√m/s); record P vs crack-mouth opening displacement V with clip gauge.
4. Draw 5 % secant line (slope 0.95 of initial slope); P_5 = intersection; **P_Q** by record type (Fig 5.2); P_{max} = max load.
5. $K_Q = (P_Q/B\sqrt{W})\ f(a/W)$; $K_Q = K_{Ic}$ only if validity criteria are met.



5 % secant offset: slope of line OA reduced 5 % → intersection gives P_5 ; $P_Q = P_5$ (or earlier max load)

Fig 5.2 Load–COD records and P_Q

Validity requirements ($K_Q \rightarrow K_{Ic}$)

> $P_{max}/P_Q \leq 1.10$ (limited non-linearity).

> **$B, a \geq 2.5 (K_Q/\sigma_{ys})^2$** (plane strain; ensures small plastic zone, 50 × r_y).

> $0.45 \leq a/W \leq 0.55$; crack front straightness (a at surface and centre within 10 %).

> If invalid → report K_Q only; use thicker specimen or EPFM (J, CTOD).

Worked Example 1 (E399 check)

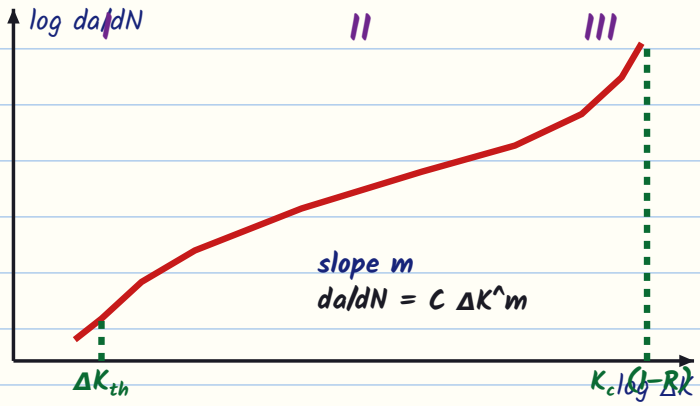
CT: W = 50 mm, B = 25 mm, a = 25.5 mm (α = 0.51), P_Q = 15.8 kN, P_{max} = 16.6 kN, σ_{ys} = 450 MPa. f(0.51) = 9.96.

$K_Q = 15800 \times 9.96/(0.025 \times \sqrt{0.05})$ = **28.2 MPa√m**. Requirement $2.5(28.2/450)^2$ = **9.8 mm**; B = 25 mm and a = 25.5 mm exceed it ✔. P_{max}/P_Q = 1.05 ≤ 1.10 ✔ ⇒ **K_{Ic} = 28.2 MPa√m** (valid).

1.2 J_{Ic} and CTOD testing

- **J_{Ic} (E1820):** single-specimen unloading-compliance (or multiple specimens); plot J vs Δa; blunting line $J = 2\sigma_Y \Delta a$; 0.2 mm offset line ⇒ J_Q; valid if B, $b_0 > 10\ J_Q/\sigma_Y$; $K_{Jc} = \sqrt{(E' J_{Ic})}$.
- **CTOD (BS 7448):** SENB with clip gauge; plastic hinge model: $\delta = K^2(1-\nu^2)/(2\sigma_{ysE}) + r_p(W-a)V_p/[r_p(W-a) + a + z]$, $r_p \approx 0.4$ (rotation factor), V_p = plastic part of V, z = knife-edge height. Reported δ_c (brittle), δ_u, δ_m.
- **Charpy** gives energy-temperature curve (DBTT), has empirical correlations with K_{Ic} (Rolfe–Barsom) – used only for screening, not for design values.

2. Paris Law and the Sigmoidal Curve



Region I : threshold, microstructure & R-sensitive, $da/dN < 10^{-9}$ m/c

Region II : Paris law (log-linear) stable, $m = 2 - 4$

Region III: $K_{max} \to K_c$, rapid, static modes, thickness-sensitive

Fig 5.3 Sigmoidal $da/dN - \Delta K$ curve (log–log)

Region	Behaviour	Controlled by
I (near threshold)	da/dN falls steeply to ~10^{-10} m/cycle at ΔK_{th}; below it no growth	microstructure, mean stress R, environment, closure
II (intermediate)	log da/dN vs log ΔK straight line: $da/dN = C \Delta K^m$	ΔK (stable, most of the life)
III (high ΔK)	da/dN accelerates as $K_{max} \to K_c$ (R) ; static fracture modes	K_{max}, K_c, thickness, microstructure

- **Paris–Erdogan (1963):** $da/dN = C (\Delta K)^m$; m = slope in log–log plot (2–4 metals, 3 ≈ steel, 3–4 Al; ceramics 10–50). C depends on units, R, environment.
- $\Delta K = K_{max} - K_{min} = \Delta\sigma\, Y \sqrt{(\pi a)}$ (for $R \geq 0$). Threshold $\Delta K_{th} \approx 6.4(1 - 0.85R)$ MPa√m for steels, ≈ 2–4 (Al) — below ΔK_{th}: infinite life (fatigue-limit analogue for cracks).

Material class (Barsom–Rolfe)	C (m/cycle, MPa√m)	m
Ferrite–pearlite steels	6.9×10^{-12}	3.0
Martensitic steels	1.35×10^{-10}	2.25
Austenitic stainless	5.6×10^{-12}	3.25
Al alloys (typical)	$\approx 5 \times 10^{-11}$	≈ 3–3.5

Worked Example 2 (Paris life)

Edge crack, $Y = 1.12$, $\Delta\sigma = \sigma_{max} = 120$ MPa ($R = 0$), $a_0 = 1$ mm, $K_c = 100$ MPa√m, ferrite-pearlite steel ($C = 6.9 \times 10^{-12}$, $m = 3$).

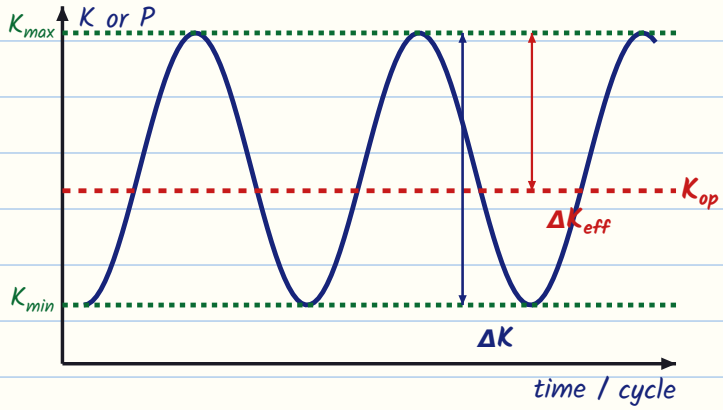
$a_c = (1/\pi)(100/(1.12 \times 120))^2 =$ **0.176 m**. $N_f = (a_c^{-0.5} - a_0^{-0.5})/[C(Y\Delta\sigma\sqrt{\pi})^3(-0.5)] = (2.382 - 31.62)/(6.9 \times 10^{-12} \times (238.2)^3 \times (-0.5)) \approx$ **6.3×10^5 cycles**.

Inspection: $N_f/4 \approx 1.6 \times 10^5$ cycles; $\Delta K_0 = 120 \times 1.12 \times \sqrt{(\pi \times 0.001)} = 7.5$ MPa√m > ΔK_{th} (5.9) ⇒ crack will grow.

2.1 Effects of R, environment, thickness

- ↑R ⇒ ↑da/dN, ↓ΔK_{th} (Region I & III most sensitive; Region II mildly).
- Corrosive environment / hydrogen / high T ⇒ faster, lower ΔK_{th}. High frequency tends to reduce environmental effect.
- Large grains, high yield & rough crack path ⇒ higher ΔK_{th} (roughness closure).

3. Crack Closure



Elber (1970):

crack faces touch (closed) before load reaches zero (wake plasticity)

$\Delta K_{eff} = K_{max} - K_{op}$

$U = \Delta K_{eff}/\Delta K = 0.5 + 0.4R$

$da/dN = C\,(U\,\Delta K)^m$

(−0.1 ≤ R ≤ 0.7, Al 2024-T3)

Fig 5.4 Elber's crack closure concept

- **Elber (1970):** fatigue cracks remain closed during part of the tensile loading because of residual plastic stretch left in the wake of the crack. Only the portion above K_{op} is effective.

$$\Delta K_{eff} = K_{max} - K_{op}\ ,\ U = \Delta K_{eff}/\Delta K = 0.5 + 0.4R\ (-0.1 \le R \le 0.7)$$

$$da/dN = C\,(\Delta K_{eff})^m = C\,(U\,\Delta K)^m$$

Closure mechanism	Cause	Where important
Plasticity-induced	residual stretch in the wake	Region II, plane stress, thin sheet
Oxide-induced	oxide debris/ corrosion product wedges crack	Region I, humid / high T, low R
Roughness-induced	mismatch of fracture surface asperities (Mode II shifts)	Region I, coarse grains, low R
Viscous-fluid / phase transformation	fluid in crack; martensite transformation zone	oils, TRIP steels, ZrO_2

Worked Example 3

$R = 0.3$, $\Delta K = 15$ MPa$\sqrt{m} \Rightarrow U = 0.5 + 0.4 \times 0.3 = 0.62 \Rightarrow \Delta K_{eff} =$ **9.3 MPa$\sqrt{m}$**; growth rate is $(0.62)^m$ times that of $\Delta K = 15$ without closure (for $m = 3 \rightarrow 24$ %).

3.1 Variable amplitude – overloads

- **Single tensile overload** (OLR = $K_{ol}/K_{max} \approx 1.5$–2): large plastic zone ⇒ compressive residual stress & increased closure ⇒ **retardation** (delay), possibly arrest at OLR ≥ 2.
- Retardation lasts until the crack grows through the overload plastic zone r_{ol}. **Underloads** and compressive loads reduce/negate retardation. Spectrum loading (rainflow cycle counting + damage summation) is used in practice.

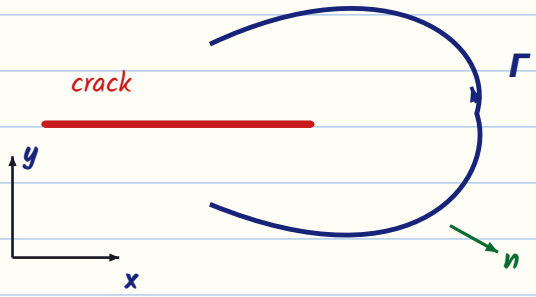
4. Crack Growth Models

Model	Equation	Remarks
Paris–Erdogan	$da/dN = C\,\Delta K^m$	Region II, constant R
Walker (1970)	$da/dN = C[\Delta K/(1-R)^{1-\gamma}]^m = C[K_{max}(1-R)^\gamma]^m$	R-effect; $\gamma \approx 0.5$–0.9 fitted
Forman (1967)	$da/dN = C\,\Delta K^m / [(1-R)K_c - \Delta K]$	includes Region III, R
Elber	$da/dN = C\,(U\,\Delta K)^m$	closure-based
Priddle / modified Forman	$da/dN = C\,\Delta K^m\,(1 - \Delta K_{th}/\Delta K)^p / (1 - K_{max}/K_c)^q$	Regions I-III (sigmoidal)
NASGRO	$da/dN = C[(1-f)/(1-R)\,\Delta K]^n\,(1-\Delta K_{th}/\Delta K)^p / (1-K_{max}/K_c)^q$	f = closure function; used in NASA/ESA codes
Wheeler (1972)	$da/dN = C_p\,(da/dN)_{CA}$; $C_p = [r_{pi}/(a_{ol} + r_{ol} - a_i)]^p$	overload retardation, p fitted
Willenborg	$K_{eff} = K_{max} - K_{comp}$ (residual stress intensity)	retardation, no fit constant (generalised)

4.1 Using the models

- Constant amplitude: Paris if one R; Walker/Forman for several R; NASGRO when sigmoidal behaviour/threshold matters (damage-tolerance design of aircraft).
- Variable amplitude: **equivalent ΔK_{rms}** = $[\Sigma\, n_i\,\Delta K_i^m / \Sigma\, n_i]^{1/m}$ or cycle-by-cycle integration with Wheeler / Willenborg / closure (FASTRAN, AFGROW, NASGRO software).
- **Wheeler example:** $r_{pi} = 0.5$ mm, overload zone $r_{ol} = 2$ mm, $a_{ol} = 10$ mm, $a_i = 10.2$ mm, $p = 1.5$: $C_p = [0.5/(10 + 2 - 10.2)]^{1.5} = (0.278)^{1.5} =$ **0.146** ⇒ growth rate reduced to 15 %.
- Life = inspection interval: with probability of detection (POD) curve & scatter factor; Retirement for cause.

5. J-Integral



Rice (1968):

$$J = \oint_\Gamma \left(w\, dy - T_i \frac{\partial u_i}{\partial x} ds \right)$$

$w = \int \sigma_{ij}\, d\varepsilon_{ij}$ (strain energy density)
$T_i = \sigma_{ij}\, n_j$ (traction on Γ)

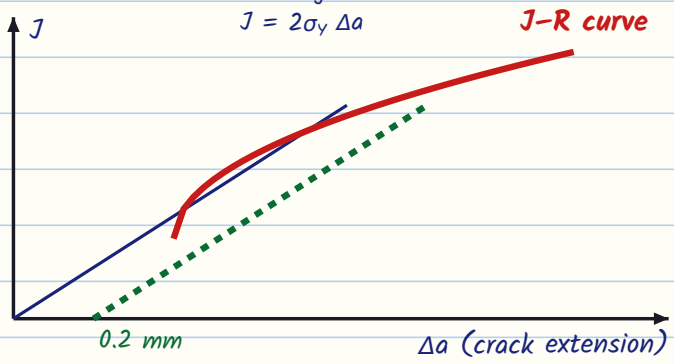
- path independent (non-linear elastic)
- $J = -d\Pi/da = G$ (linear elastic)
- $J = 0$ on closed path w/o singularity

Fig 5.5 Contour for the J-integral

- For a **non-linear elastic** material (deformation plasticity), J is path independent. Physical meaning: **$J = -(1/B)\, d\Pi/da$** — energy release rate for non-linear elastic bodies. Linear elastic: **$J = G = K^2/E'$**.
- HRR field (Hutchinson, Rice, Rosengren) for Ramberg–Osgood material $\varepsilon/\varepsilon_0 = \sigma/\sigma_0 + \alpha(\sigma/\sigma_0)^n$:

$$\sigma_{ij} = \sigma_0 \left[J / (\alpha\, \varepsilon_0\, \sigma_0\, I_n\, r) \right]^{1/(n+1)} \tilde{\sigma}_{ij}(\theta, n)$$

- J **uniquely** characterises the stress/strain singularity ($\sigma \sim r^{-1/(n+1)}$, $\varepsilon \sim r^{-n/(n+1)}$); $n = 1$ recovers LEFM ($r^{-1/2}$). Fracture at $J = J_{Ic}$.
- **Relations:** $\delta_t = J/(m_1 \sigma_{ys})$ ($m_1 \approx 1$–2.5); $K_{Jc} = \sqrt{(E' J_c)}$; $J = J_{el} + J_{pl}$; $J_{el} = K^2/E'$.



J_{Ic} (ASTM E1820):

intersection J_Q of R-curve with 0.2 mm offset line (parallel to blunting line)

valid if $B, b_0 > 10\, J_Q/\sigma_Y$

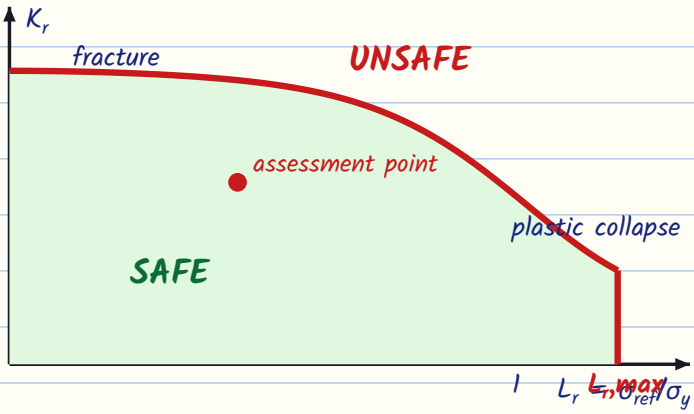
Tearing modulus $T = (E/\sigma_0^2)\, dJ/da$
Stable growth: $T_{app} < T_{mat}$

Fig 5.6 J–R curve and determination of J_{Ic}

- **EPRI estimation:** $J = J_e(a_{eff}) + \alpha\, \varepsilon_0\, \sigma_0\, b\, (a/W)\, h_1(a/W, n)\, (P/P_0)^{n+1}$ (b = ligament, P_0 = limit load).
- Limitations: no unloading (deformation theory), small crack growth $\Delta a \ll b \Rightarrow$ J-controlled growth $\omega = (b/J)(dJ/da) \gg 1$.

Example: $K = 60$ MPa$\sqrt{m}$ (plane strain, $E = 210$ GPa, $\nu = 0.3$): $J = (1-\nu^2)K^2/E = 0.91 \times (60\times10^6)^2/(210\times10^9) =$ **15.6 kJ/m²**; $\delta = J/(2\sigma_{ys}) = 15600/(2\times400\times10^6) =$ **0.0195 mm**.

6. Failure Assessment Diagram (FAD)



Option 1 (BS 7910 / R6):

$K_r = K_I/K_{mat}$ (+ρ)

$L_r = \sigma_{ref}/\sigma_y = P/P_L$

$f(L_r) = (1-0.14L_r^2)$ × $[0.3 + 0.7\ \exp(-0.65\ L_r^6)]$

cut-off $L_{r,max} = (\sigma_y+\sigma_u)/2\sigma_y$

Fig 5.7 FAD – two-criteria approach (R6 / BS 7910 Option 1)

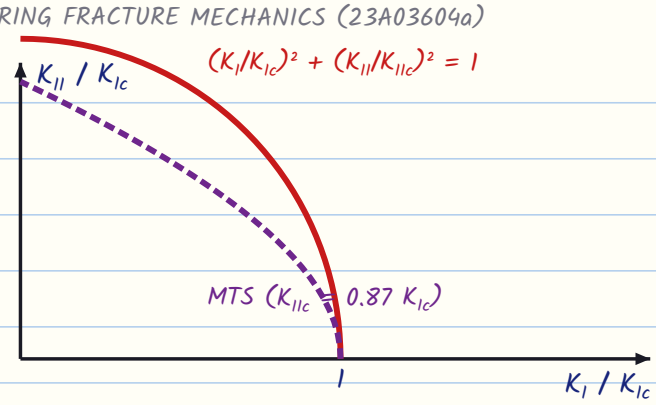
- Interpolates between **brittle fracture (K_r = 1)** and **plastic collapse (L_r = 1)**. Components with flaws are assessed by a single point (L_r, K_r).

1. Characterise flaw (a, shape) from NDT; apply conservative flaw-size interaction/ recategorisation rules.
2. Calculate K_I (primary + secondary stresses, with plasticity correction ρ) ⇒ **$K_r = K_I/K_{mat}$** ($K_{mat} = K_{Ic}$, or from J/δ).
3. Reference stress σ_{ref} (net-section) ⇒ **$L_r = \sigma_{ref}/\sigma_y = P/P_L$**.
4. Plot (L_r, K_r). Inside the failure line $K_r = f(L_r)$ and $L_r < L_{r,max}$ ⇒ **acceptable**; on/outside ⇒ unacceptable (repair / re-rate / reduce load).

Example: K_I = 50, K_{mat} = 100 MPa√m ⇒ K_r = 0.50; σ_{ref} = 240, σ_y = 400 ⇒ L_r = 0.60. $f(0.6) = 0.9496 \times 0.979 = 0.93 > 0.50$ ⇒ **safe**. Reserve factor along the ray ≈ 1.46 (load can rise ~46 % before the assessment line is reached).

- Cut-off $L_{r,max} = (\sigma_y + \sigma_u)/(2\sigma_y)$ (e.g. 1.2 for σ_u/σ_y = 1.4). Option 2 & 3 use actual stress–strain curve / J-based FAD (more accurate, less conservative).
- Used in BS 7910, API 579, R6, SINTAP, FITNET: fitness-for-service of pressure vessels, pipelines, nuclear and offshore structures.

7. Mixed-Mode Fracture



Mixed-mode envelopes (I + II)

Inside the curve : no fracture
Outside / on curve : fracture

Kink angle (MTS):

$\theta_0 = 2\tan^{-1}[(K_I - \sqrt{(K_I^2+8K_{II}^2)})/4K_{II}]$

pure II : $\theta_0 = -70.5°$

Fig 5.8 Mixed-mode I + II failure envelopes

Criterion	Statement	Result for pure Mode II
Max tangential stress (MTS, Erdogan-Sih)	Crack grows ⊥ to max $\sigma_{\theta\theta}$; at θ_0 : $K_I \sin\theta_0 + K_{II}(3\cos\theta_0 - 1) = 0$; fracture when $\cos(\theta_0/2)[K_I \cos^2(\theta_0/2) - 1.5K_{II}\sin\theta_0] = K_{Ic}$	$\theta_0 = -70.5°$, $K_{IIc} = 0.87 K_{Ic}$
Strain-energy density (SED, Sih)	Growth toward S_{min} ; fracture when $S_{min} = S_c$	$K_{IIc} \approx 1.05\ K_{Ic}$ ($\nu = 0.3$)
Max energy release rate (Hussain)	Growth in direction of G_{max} for kinked crack	$\theta_0 \approx -75°$, $K_{IIc} = K_{Ic}$
Simple interaction	$(K_I/K_{Ic})^2 + (K_{II}/K_{IIc})^2 = 1$; or $G_I + G_{II} = G_c$	—

- Equivalent K: $K_{eq} = \sqrt{(K_I^2 + K_{II}^2)}$ (conservative, from $G = G_I + G_{II}$), or MTS-based K_{eq} above. Mode III adds $K_{III}^2(1+\nu)$.

Example (Unit IV, Ex.1): $K_I = 13.29$, $K_{II} = 7.67 \Rightarrow \theta_0 = -43.2°$; $K_{eq}(MTS) = \cos(\theta_0/2)[K_I \cos^2(\theta_0/2) - 1.5K_{II}\sin\theta_0] \approx$ **18.0 MPa√m** (vs $\sqrt{(K_I^2+K_{II}^2)} = 15.3$). Fracture if $K_{eq} \geq K_{Ic}$.

- Experiments: inclined centre-cracked plate, CTS (Richard) specimen, Arcan fixture, 4-point-bend SENB; fatigue growth of mixed-mode cracks: ΔK_{eq}-Paris, crack kinks and then grows in near Mode I.

8. Crack Arrest and Repair

8.1 Crack arrest

- A running (dynamic) crack stops when the dynamic driving force K_I falls below the **arrest toughness K_{Ia}** ($< K_{Ic}$). Approaches: tough steel (low DBTT), lower stress regions, **crack arrester** strips/ stiffener, riveted seams (Liberty-ship lessons), tough inserts in pipelines, crack arrest rings.
- Tests: Robertson test (crack arrest temperature CAT), ESSO test, Double-cantilever (K_{Ia}, ASTM E1221), drop-weight NDT temperature (E208). Fail-safe: rivet/ stringer-frame arrangement limits K as crack approaches a stiffener (K falls).

8.2 Repair methodologies

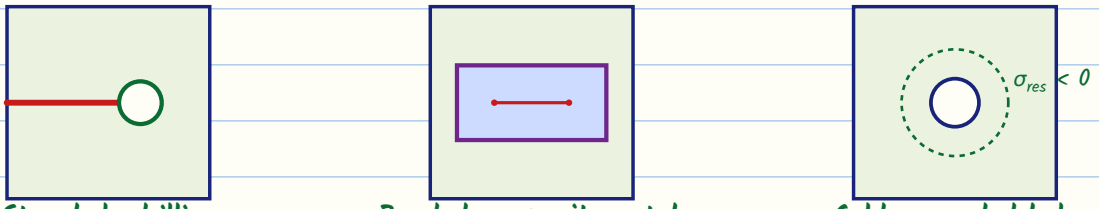


Fig 5.9 Typical repair techniques

Method	Principle / remarks
Stop-hole drilling	Drill hole (radius ρ) just ahead of tip: blunts crack, $K_t \approx 1 + 2\sqrt{(a/\rho)}$. Temporary; cold-work or insert fastener to extend life.
Grind-out & re-weld	Remove crack fully (verify by DPT/MPI/UT), V-groove, weld with low-hydrogen electrodes, PWHT / peen.
Bonded composite patch (boron/epoxy, CFRP)	Bridges crack, reduces ΔK 50-90 %, little added stress; used on aircraft (F-111, RAAF), pipelines.
Cold working / mandrel expansion	Residual compressive stress around hole (2–6 % expansion) ⇒ fatigue life ×3–10.
Shot / laser / hammer peening, HFMI	Surface compressive stress; delays initiation & growth at weld toes.
Reinforcement: doublers, sleeves, clamps	Reduce local stress; pipeline composite wrap, Type B sleeve.
Stress relief / PWHT	Reduces tensile residual stress (K_{res}).

- **Inspection** (NDT): dye penetrant, magnetic particle, eddy current, ultrasonic (TOFD, phased array), radiography. Damage-tolerance plan: $a_{detectable} \rightarrow a_c$; interval ≤ N/(scatter factor 2–4).
- **Repair decision:** assess by FAD / Paris life; if not acceptable → repair, derate load, or retire.

9A. Additional Notes & Examples

A. Fatigue crack growth testing (ASTM E647)

1. CT or M(T) specimen, fatigue pre-crack to a_0 at the test ΔK; constant-amplitude load, R fixed (0.1 typical), f = 10–30 Hz.
2. Monitor a: travelling microscope, compliance (clip gauge), potential drop (DCPD) – record (a, N) pairs.
3. da/dN by **secant** ($\Delta a/\Delta N$ between points) or **7-point incremental polynomial**; ΔK from handbook $Y(a/W)$.
4. Near threshold: **load-shedding** (ΔK-decreasing, $C = (1/\Delta K)(d\Delta K/da) \approx -0.08\ mm^{-1}$) until da/dN ~ 10^{-10} m/cycle $\Rightarrow \Delta K_{th}$.
5. Plot log da/dN vs log ΔK $\Rightarrow$ fit C, m (Region II), ΔK_{th}, K_c.

B. Numerical – effect of R (Walker vs Paris)

Walker with $C = 3\times10^{-12}$, m = 3, γ = 0.5: for ΔK = 20 MPa$\sqrt{m}$ at R = 0.5, $\Delta K/(1-R)^{0.5}$ = 28.3 $\Rightarrow$ da/dN = $3\times10^{-12} \times 28.3^3$ = **6.8×10^{-8} m/cycle**; at R = 0 : $3\times10^{-12}\times20^3 = 2.4\times10^{-8}$.

Ratio 2.83 $\Rightarrow$ the mean-stress effect nearly triples the growth rate.

Forman (illustrative): $C = 3\times10^{-10}$, m = 3, K_c = 60, R = 0.5, ΔK = 20 $\Rightarrow$ da/dN = $3\times10^{-10}\times8000/(0.5\times60 - 20)$ = **2.4×10^{-7}**; as $\Delta K \to (1-R)K_c$ = 30 the denominator $\to$ 0 and da/dN $\to \infty$ (Region III).

C. Tearing modulus (ductile stable growth)

- $T_{mat} = (E/\sigma_0^2)(dJ/da)$. Stable growth requires $T_{app} < T_{mat}$; instability when the applied tearing modulus $T_{app} = (E/\sigma_0^2)(\partial J/\partial a)_\Delta$ equals T_{mat}.

Example: E = 210 GPa, σ_0 = 400 MPa, dJ/da = 200 MN/m² $\Rightarrow$ $T = (210\times10^9/(400\times10^6)^2) \times 2\times10^8$ = **263** (typical for tough structural steels: 50–500).

D. FAD – further points

- **Secondary stresses** (residual, thermal) raise K but do not contribute to plastic collapse: $K_r = (K_p + K_s)/K_{mat} + \rho$, where ρ is a plasticity interaction term (small at low L_r).
- **Reference stress** examples: $\sigma_{ref} = P/(B(W-a))$ × factor for CT; for a through crack in a plate $\sigma_{ref} = \sigma_{net}$-section ($\sigma$ W/(W − 2a)).
- A flaw that is unacceptable may be accepted after more refined analysis (Option 2/3), better K_{mat} (CTOD/J data) or removing conservatism (actual σ_y).

E. Two-mark answers

Q. Why is fatigue pre-cracking required? To obtain a natural, sharp crack (root radius $\to$ 0) so measured toughness is a lower-bound material property.

Q. What is ΔK_{th}? Stress-intensity range below which fatigue cracks do not propagate (da/dN < 10^{-10} m/cycle).

Q. State two limitations of Paris law. Valid only for Region II; ignores R, threshold, Region III and load-sequence effects.

Q. What is J in linear elasticity? $J = G = K^2/E'$.

Q. What does a point outside the FAD curve mean? The flaw is unacceptable: failure by fracture/collapse is predicted.

Q. Name two crack-repair methods that induce compressive stress. Cold-expansion of holes and peening.

Q. What is retardation? Temporary reduction of crack growth rate after a tensile overload.

9B. Damage-Tolerance Design Example

A. Residual-strength curve

Edge crack, $Y = 1.12$, $K_c = 100$ MPa$\sqrt{m}$: $\sigma_f = K_c/(Y\sqrt{(\pi a)})$. (Capped by yield/net-section collapse, e.g. 400 MPa.)

a (mm)	Failure stress σ_f (MPa)
5	712
10	504
20	356
50	225
100	159

- Strength falls rapidly as the crack grows; at $a = 50$ mm the part fails at only 225 MPa. Small cracks ($a \leq 10$ mm) are limited by plastic collapse rather than fracture.

B. Inspection interval

Same steel as Example 2 ($C = 6.9\times10^{-12}$, $m = 3$, $\Delta\sigma = 120$ MPa, $Y = 1.12$). NDT can reliably detect $a_d = 2$ mm. Life from 2 mm to $a_c = 176$ mm: $N = (0.176^{-0.5} - 0.002^{-0.5})/(C(238.2)^3(-0.5))$ = **4.3e+05 cycles**.

With scatter factor 2 ⇒ inspect every ≈ **2.1e+05 cycles** (e.g. ~ 2 years at 10^5 cycles/year). If a crack ≥ a_d is found ⇒ repair (stop-hole/ weld/ patch) or replace.

C. Stop-hole numerical

Crack $a = 20$ mm stopped by a hole of radius $\rho = 3$ mm: $K_t \approx 1 + 2\sqrt{(a/\rho)} = 1 + 2\sqrt{(20/3)}$ = **6.2** (vs ∞ for a sharp crack). Larger hole radius ⇒ lower K_t but more stress concentration due to hole itself; typical hole diameter ≈ 3–6 × plate thickness for thin sheet, drilled so that the crack tip lies within the hole.

D. Comparison summary of fracture parameters

Parameter	Applicability	Criterion	Units
K	LEFM, SSY	$K = K_{Ic}$	MPa√m
G	LEFM	$G = G_c$	J/m² (N/m)
CTOD δ	LEFM → EPFM	$\delta = \delta_c$	mm
J	EPFM (non-linear elastic)	$J = J_{Ic}$, $T_{app} < T_{mat}$	kJ/m²
FAD (K_r, L_r)	Engineering assessment	(L_r, K_r) inside curve	—

9. Exam Questions & Formula Sheet (Unit V)

1. Describe ASTM E399 K_{Ic} test: specimen, pre-cracking, P_Q determination and validity criteria. Solve a numerical.
2. Sketch the sigmoidal da/dN–ΔK curve and explain the three regions. State Paris law and estimate life.
3. Explain crack closure (Elber) and its effect on growth; discuss overload retardation.
4. Write Walker, Forman, NASGRO and Wheeler models and state their use.
5. Define J-integral; show path independence idea; state HRR field, J_{Ic} determination and J–CTOD relation.
6. Explain FAD and procedure for assessing a flaw. Solve a numerical.
7. Explain mixed-mode criteria (MTS, SED, G_{max}). Find the kink angle.
8. Discuss crack arrest and repair methods with examples.

Formula sheet – Unit V

> K_{Ic} valid: $B, a \geq 2.5(K_Q/\sigma_{ys})^2$; $P_{max}/P_Q \leq 1.1$; $K_Q = (P_Q/B\sqrt{W})\, f(a/W)$

> Paris: $da/dN = C\,\Delta K^m$; $N = [a_c^{1-m/2} - a_0^{1-m/2}] / [C(Y\Delta\sigma\sqrt{\pi})^m(1-m/2)]$

> Closure: $\Delta K_{eff} = K_{max} - K_{op}$; $U = 0.5 + 0.4R$; Walker, Forman: see §4

> $J = -d\Pi/da$; $J = K^2/E'$ (LEFM) ; $\delta = J/(m\sigma_{ys})$; HRR $\sigma \sim r^{-1/(n+1)}$

> FAD: $K_r = K_I/K_{mat}$; $L_r = \sigma_{ref}/\sigma_y$; $f = (1-0.14L_r^2)[0.3+0.7e^{-0.65L_r^6}]$

> MTS: $\theta_0 = 2\tan^{-1}[(K_I - \sqrt{(K_I^2+8K_{II}^2)})/4K_{II}]$